

24IMCD2P1

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June 5, 2026

Problem

Prove that for any function $f : \mathbb{Q} \rightarrow \mathbb{Z}$, there exist $a, b, c \in \mathbb{Q}$ such that $a < b < c$, $f(b) \geq f(a)$ and $f(b) \geq f(c)$.

Solution. Let $a < c_1$ be two arbitrary real numbers and say $f(a) \leq f(c_1)$ wolog. Define $c_k = \frac{a + c_{k-1}}{2}$ for all $k \geq 1$.

Since there are only finitely many integers between $f(a)$ and $f(c_1)$, for some k we must have $\min(f(a), f(c_{k-1})) \geq f(c_k)$.

If for any $b \in (a, c_{k-1})$ we have $f(b) \geq \max(f(a), f(c_{k-1}))$ then we are done. Assume otherwise. Observe that if $\exists q_1, q_2$ such that $q_1 \in (a, c_k), q_2 \in (c_k, c_{k-1})$ such that $\max(f(q_1), f(q_2)) \leq f(c_k)$ then we are done. If the contrary holds true then there is an $I \in \{(a, c_k), (c_k, c_{k-1})\}$ such that $\forall q \in I, f(c_k) \leq f(q) \leq \max(f(a), f(c_{k-1}))$. Since there are only finitely many integers in between $f(a)$ and $f(c_{k-1})$, one value must be attained infinitely many times, say t , then we must have x_1, x_2, x_3 such that $f(x_1) = f(x_2) = f(x_3)$, proving the statement.